

Practical Demo: Hedging a Bond Portfolio with Treasury Futures

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

How many Treasury futures offset the portfolio's rate exposure, and which losses remain after that hedge?

Learning Objectives

After completing this demo you should be able to:

- calculate a bond portfolio's analytical and Treasury-beta-adjusted DV01;
- derive Treasury futures DV01 from the cheapest-to-deliver (CTD) bond and conversion factor;
- size and round a short futures hedge;
- compare portfolio P&L before and after the hedge; and
- identify residual basis, spread, curve, rounding, and CTD risks.

1 Market Inputs Versus Illustrative Inputs

The desk request: reduce the portfolio's parallel Treasury-rate risk without selling its corporate bonds. The workflow is **holdings** → **Treasury DV01** → **CTD mapping** → **integer hedge** → **residual P&L**. Equal face amounts do not imply equal exposures. The hedge is judged against the specified risk, not against whether the entire portfolio avoids a loss.

The workflow resembles a daily risk-management task, but every number below is illustrative rather than live market data. A live hedge would use current positions, clean prices, accrued interest, key-rate durations, deliverable-basket data, conversion factors, CTD candidates, financing rates, and exchange contract specifications.

Simplifying assumptions:

- changes are small enough for first-order DV01 approximations;
- the selected CTD remains unchanged;
- futures DV01 is constant over the scenario;
- the Treasury move is parallel;
- spread duration equals modified duration for the corporate positions;
- yield betas summarize how each position historically responds to Treasury yields; and
- transaction costs, margin cash flows, convexity, carry, and roll are ignored.

Workflow stage	In this case
Observed or supplied	Positions, CTD price, conversion factor, and futures price
Estimated	Treasury yield betas and CTD-based futures DV01
Solved	Integer number of contracts to short
Diagnosed	Residual rate DV01, spread risk, and rounding error
Used for	Before-and-after hedge P&L under a joint rate/spread shock

2 Build the Portfolio Risk Table

The portfolio contains Treasuries and investment-grade corporate bonds. Analytical DV01 is

$$DV01_i = D_{\text{mod},i} \times MV_i \times 0.0001.$$

For cross-hedging, a Treasury yield beta maps each bond's historical price sensitivity to the hedging instrument:

$$DV01_i^{\text{Treasury}} = DV01_i \times \beta_i.$$

Position	5-year Treasury notes	10-year Treasury notes	A industrial bonds
Market value (\$mm)	30	25	24
Modified duration	4.5	7.6	6.1
Treasury yield beta	1	1	0.88
Spread shock in scenario (bp)	0	0	12
Analytical DV01 (\$/bp)	13,500	19,000	14,640
Treasury DV01 (\$/bp)	13,500	19,000	12,883.2
Spread DV01 (\$/bp)	0	0	14,640

Position	BBB financial bonds
Market value (\$mm)	18
Modified duration	7.2
Treasury yield beta	0.72

Position	BBB financial bonds
Spread shock in scenario (bp)	20
Analytical DV01 (\$/bp)	12,960
Treasury DV01 (\$/bp)	9,331.2
Spread DV01 (\$/bp)	12,960

```

Portfolio market value ($mm)          97.0
Analytical DV01 ($/bp)                60100.0
Treasury-beta-adjusted DV01 ($/bp)    54714.4
Corporate spread DV01 ($/bp)          27600.0
dtype: float64

```

The analytical DV01 describes a common change in every position's discount yield. The beta-adjusted DV01 is the exposure targeted by a Treasury futures hedge. Spread risk remains a separate exposure.

3 Frame the CTD and Conversion-Factor Problem

Treasury futures are linked to a delivery basket. The short chooses the eligible security to deliver, and the cheapest-to-deliver issue often determines the futures contract's effective interest-rate sensitivity.

Suppose the current CTD candidate has the following illustrative inputs:

Input	Value
CTD clean price	101.75
CTD modified duration	7.35
Deliverable face amount	\$100,000
Conversion factor	0.9275
Futures price	112-16 = 112.50

Ignoring accrued interest and delivery-option details,

$$DV01_{CTD} = D_{\text{mod},CTD} P_{CTD}(0.0001),$$

and

$$DV01_{futures} \approx \frac{DV01_{CTD}}{CF}.$$

CTD market value (\$)	101750.00
Approximate invoice price before accrued interest (\$)	104343.75
CTD DV01 (\$/bp)	74.79
Futures DV01 (\$/bp)	80.63
dtype: float64	

The futures contract does not have a true bond duration. Its practical sensitivity is inherited from the current CTD bond and scaled by the conversion factor.

4 Size and Round the Hedge

i One contract, with units

The CTD's market value is $\$100,000 \times 101.75 / 100 = \$101,750$. Multiplying by 7.35 years and 0.0001 gives about **\$74.79 per bp**. Dividing by the 0.9275 conversion factor gives about **\$80.63 per bp per futures contract**. Portfolio dollars per bp divided by contract dollars per bp therefore gives a number of contracts—not a dollar notional.

The portfolio is long bonds and loses when rates rise, so the hedge shorts Treasury futures. The exact contract count is

$$N^* = \frac{DV01_{portfolio}^{Treasury}}{DV01_{futures}}.$$

Exact contracts to short	678.5687
Rounded contracts to short	679.0000
Hedge DV01 (\$/bp)	54749.1792
Residual Treasury DV01 (\$/bp)	-34.7792
Hedge ratio	1.0006
dtype: float64	

Rounding creates a small residual exposure. A negative residual means the rounded hedge is slightly too large; a positive residual means it is slightly too small.

5 Before-and-After Scenario P&L

Suppose Treasury yields rise by 35 basis points. Corporate spreads also widen by the amounts in the portfolio table. The first-order bond P&L is separated into Treasury and spread components:

$$\Delta V_{portfolio} \approx -DV01^{Treasury} \Delta y_{Treasury} - DV01^{spread} \Delta s.$$

Because the hedge is short futures, its approximate gain when Treasury yields rise is

$$\Delta V_{futures} \approx N \times DV01_{futures} \times \Delta y_{Treasury}.$$

Position	Treasury P&L (\$)	Spread P&L (\$)	Total unhedged P&L (\$)
5-year Treasury notes	-472,500	-0	-472,500
10-year Treasury notes	-665,000	-0	-665,000
A industrial bonds	-450,912	-175,680	-626,592
BBB financial bonds	-326,592	-259,200	-585,792

Unhedged portfolio P&L (\$)	-2349884.0
Short futures P&L (\$)	1916221.0
Hedged portfolio P&L (\$)	-433663.0
Residual rate P&L from rounding (\$)	1217.0
Unhedged spread P&L (\$)	-434880.0
dtype: float64	

The futures hedge removes almost all of the modeled Treasury-rate loss. It does not hedge the corporate spread widening, which remains the dominant loss after the hedge.

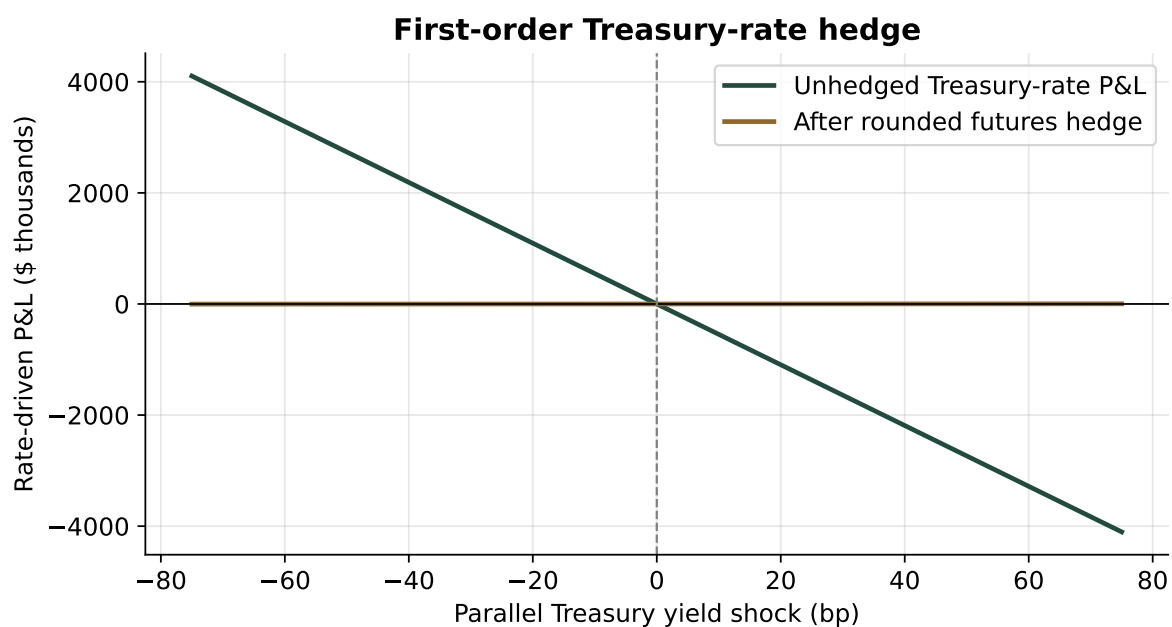


Figure 1: The futures hedge offsets parallel Treasury-rate risk but not spread risk

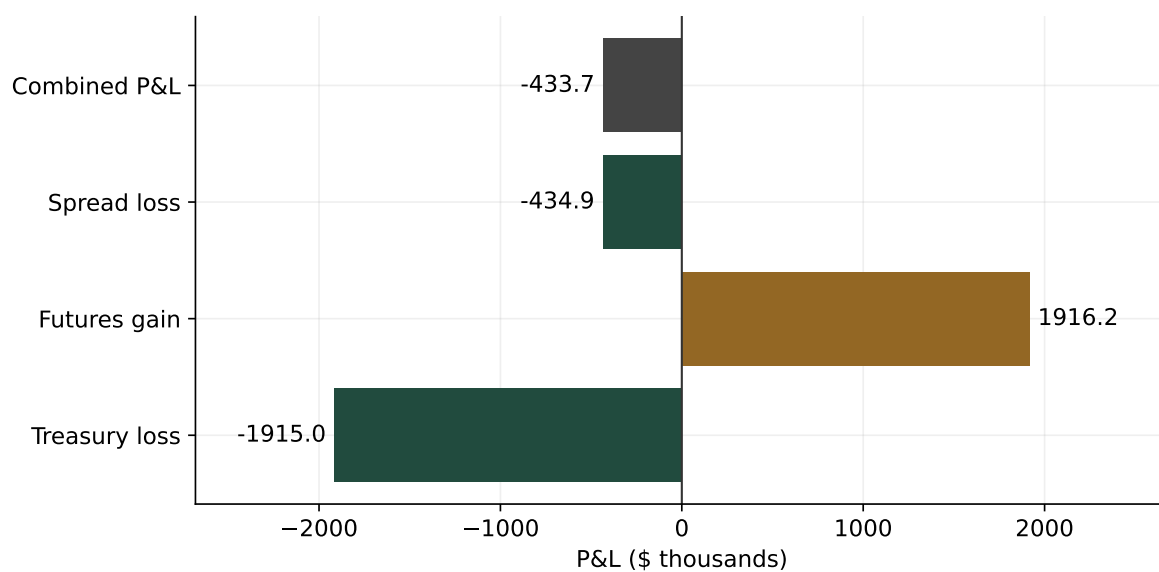


Figure 2: The rate loss and futures gain largely offset; the corporate-spread loss remains.

6 Validation Checks

6.1 Reconcile what the hedge actually removed

Change the hedge objective

Before rerunning, predict the result of hedging only 50% of Treasury DV01. Replace the numerator of the contract-count calculation with half the exposure. The spread loss should stay unchanged; the rate loss should be roughly halved. Then increase the assumed futures DV01 by 10% to represent a different CTD mapping. Keeping the old contract count would overhedge.

Futures DV01 is positive	True
Long-bond hedge requires short futures	True
Rounded residual is no more than half one contract DV01	True
Hedged P&L reconciles to spread plus residual-rate P&L	True

dtype: bool

7 What Can Break the Hedge?

- **Basis risk:** cash bonds and the futures contract may not move together.
- **Curve risk:** one CTD-based DV01 cannot hedge every key-rate exposure.
- **Spread risk:** Treasury futures do not directly hedge corporate credit spreads.
- **Yield-beta error:** historical betas can change in stress periods.
- **CTD switching:** a different deliverable bond can become cheapest to deliver.
- **Conversion-factor and delivery-option effects:** the simple DV01 scaling is approximate.
- **Convexity:** large yield moves make a first-order DV01 hedge less accurate.
- **Rounding, liquidity, margin, and roll:** implementation affects realized results.

8 Financial Interpretation

A Treasury futures hedge is a risk-matching exercise, not a notional-matching exercise. The correct sequence is:

1. identify the portfolio risk to hedge;
2. map that risk into Treasury DV01;
3. estimate futures DV01 from the current CTD and conversion factor;
4. choose and round the contract count; and

5. test the combined position under rate, spread, curve, and basis scenarios.

The hedge is successful here because it neutralizes the stated parallel Treasury-rate exposure. The remaining loss is economically meaningful: it is primarily credit spread risk, which was never within the hedge's scope.

8.1 Market handoff

[CME's Treasury hedging guide](#) explains CTD-based sensitivity matching. For a live case, record the exact delivery month, eligible basket, CTD assumption, conversion factor, and observation time. The embedded positions and CTD are a complete illustrative fallback, not downloaded quotes. The separate spread shock here is the **residual** spread move beyond any co-movement already represented by the assumed yield beta; otherwise the scenario could count that co-movement twice.

Your deliverable: a hedge ticket stating direction, integer contracts, residual DV01, stress P&L, and the unhedged risk that matters most.

9 Interactive explorer

Choose how much of Treasury DV01 to hedge in the +35 bp scenario. Contract counts are rounded at every setting. At 100%, spread losses remain. Above 100%, the position becomes net short the modeled Treasury exposure; it is no longer simply reducing that exposure.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

10 Mini Project

Project brief. Reduce the portfolio's beta-adjusted Treasury DV01 by 60%, leaving 40% in place. Report the exact and rounded short-futures count, residual Treasury DV01, and P&L under +25 and -25 bp Treasury moves combined with +40 bp corporate spread widening. Use the demo's unchanged CTD and conversion factor. Explain whether the hedge meets its objective and what a CTD switch could change.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

Exact short contracts	407.1412
Rounded short contracts	407.0000
Residual Treasury DV01	21897.1453
Residual fraction	0.4002

dtype: float64

	Treasury shock (bp)	Cash rate P&L	Futures P&L
0	-25	1,367,860	-820,431.37
1	25	-1,367,860	820,431.37

	Spread P&L	Net P&L
0	-1,104,000	-556,571.37
1	-1,104,000	-1,651,428.63

Short futures offset Treasury-rate losses but also give up part of the gain in a rally. The remaining rate exposure is 40% of the original, up to half a contract's DV01 from rounding. Corporate spread widening remains unhedged in both scenarios. A CTD switch or a change in yield beta changes effective futures or portfolio DV01 and requires resizing. Basis, convexity, daily margin cash flows, and transaction costs are omitted.